

A Technical Introduction to Question answering

Section 1

Mathematics An open-source, math-aware, question answering system called MathQA, based on Ask Platypus and Wikidata, was published in 2018. MathQA takes an English or Hindi natural language question as input and returns a mathematical formula retrieved from Wikidata as a succinct answer, translated into a computable form that allows the user to insert values for the variables. The system retrieves names and values of variables and common constants from Wikidata if those are available. It is claimed that the system outperforms a commercial computational mathematical knowledge engine on a test set. MathQA is hosted by Wikimedia at <https://mathqa.wmflabs.org/>. In 2022, it was extended to answer 15 math question types. MathQA methods need to combine natural and formula language. One possible approach is to perform supervised annotation via Entity Linking. The "ARQMath Task" at CLEF 2020 was launched to address the problem of linking newly posted questions from the platform Math Stack Exchange to existing ones that were already answered by the community. Providing hyperlinks to already answered, semantically related questions helps users to get answers earlier but is a challenging problem because semantic relatedness is not trivial. The lab was motivated by the fact that 20% of mathematical queries in general-purpose search engines are expressed as well-formed questions. The challenge contained two separate sub-tasks. Task 1: "Answer retrieval" matching old post answers to newly posed questions, and Task 2: "Formula retrieval" matching old post formulae to new questions. Starting with the domain of mathematics, which involves formula language, the goal is to later extend the task to other domains (e.g., STEM disciplines, such as chemistry, biology, etc.), which employ other types of special notation (e.g., chemical formulae). The inverse of mathematical question answering—mathematical question generation—has also been researched. The PhysWikiQuiz physics question generation and test engine retrieves mathematical formulae from Wikidata together with semantic information about their constituting identifiers (names and values of variables). The formulae are then rearranged to generate a set of formula variants. Subsequently, the variables are substituted with random values to generate a large number of different questions suitable for individual student tests.

Section 2

In information retrieval, an open-domain question answering system tries to return an answer in response to the user's question. The returned answer is in the form of short texts rather than a list of relevant documents. The

system finds answers by using a combination of techniques from computational linguistics, information retrieval, and knowledge representation. The system takes a natural language question as an input rather than a set of keywords, for example: "When is the national day of China?" It then transforms this input sentence into a query in its logical form. Accepting natural language questions makes the system more user-friendly, but harder to implement, as there are a variety of question types and the system will have to identify the correct one in order to give a sensible answer. Assigning a question type to the question is a crucial task; the entire answer extraction process relies on finding the correct question type and hence the correct answer type. Keyword extraction is the first step in identifying the input question type. In some cases, words clearly indicate the question type, e.g., "Who", "Where", "When", or "How many"—these words might suggest to the system that the answers should be of type "Person", "Location", "Date", or "Number", respectively. POS (part-of-speech) tagging and syntactic parsing techniques can also determine the answer type. In the example above, the subject is "Chinese National Day", the predicate is "is" and the adverbial modifier is "when", therefore the answer type is "Date". Unfortunately, some interrogative words like "Which", "What", or "How" do not correspond to unambiguous answer types: Each can represent more than one type. In situations like this, other words in the question need to be considered. A lexical dictionary such as WordNet can be used for understanding the context. Once the system identifies the question type, it uses an information retrieval system to find a set of documents that contain the correct keywords. A tagger and NP/Verb Group chunker can verify whether the correct entities and relations are mentioned in the found documents. For questions such as "Who" or "Where", a named-entity recogniser finds relevant "Person" and "Location" names from the retrieved documents. Only the relevant paragraphs are selected for ranking. A vector space model can classify the candidate answers. Check if the answer is of the correct type as determined in the question type analysis stage. An inference technique can validate the candidate answers. A score is then given to each of these candidates according to the number of question words it contains and how close these words are to the candidate—the more and the closer the better. The answer is then translated by parsing into a compact and meaningful representation. In the previous example, the expected output answer is "1st Oct."

Section 3

History Two early question answering systems were BASEBALL and LUNAR. BASEBALL answered questions about Major League Baseball over a period of one year. LUNAR answered questions about the geological analysis of rocks returned by the Apollo Moon missions. Both question answering systems were very effective in their

chosen domains. LUNAR was demonstrated at a lunar science convention in 1971 and it was able to answer 90% of the questions in its domain that were posed by people untrained on the system. Further restricted-domain question answering systems were developed in the following years. The common feature of all these systems is that they had a core database or knowledge system that was hand-written by experts of the chosen domain. The language abilities of BASEBALL and LUNAR used techniques similar to ELIZA and DOCTOR, the first chatterbot programs. SHRDLU was a successful question-answering program developed by Terry Winograd in the late 1960s and early 1970s. It simulated the operation of a robot in a toy world (the "blocks world"), and it offered the possibility of asking the robot questions about the state of the world. The strength of this system was the choice of a very specific domain and a very simple world with rules of physics that were easy to encode in a computer program. In the 1970s, knowledge bases were developed that targeted narrower domains of knowledge. The question answering systems developed to interface with these expert systems produced more repeatable and valid responses to questions within an area of knowledge. These expert systems closely resembled modern question answering systems except in their internal architecture. Expert systems rely heavily on expert-constructed and organized knowledge bases, whereas many modern question answering systems rely on statistical processing of a large, unstructured, natural language text corpus. The 1970s and 1980s saw the development of comprehensive theories in computational linguistics, which led to the development of ambitious projects in text comprehension and question answering. One example was the Unix Consultant (UC), developed by Robert Wilensky at U.C. Berkeley in the late 1980s. The system answered questions pertaining to the Unix operating system. It had a comprehensive, hand-crafted knowledge base of its domain, and it aimed at phrasing the answer to accommodate various types of users. Another project was LILOG, a text-understanding system that operated on the domain of tourism information in a German city. The systems developed in the UC and LILOG projects never went past the stage of simple demonstrations, but they helped the development of theories on computational linguistics and reasoning. Specialized natural-language question answering systems have been developed, such as EAGLi for health and life scientists. Question answering systems have been extended in recent years to encompass additional domains of knowledge. For example, systems have been developed to automatically answer temporal and geospatial questions, questions of definition and terminology, biographical questions, multilingual questions, and questions about the content of audio, images, and video. Current question answering research topics include: